

Shraddha Krishnavallabhdas Meshram

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EDUCATION

MS in Computer Software Engineering, Northeastern University - Boston, MA Expected Dec 2025
BE in Electrical Engineering, SRCOEM - Nagpur, India May 2022

EXPERIENCE

Noise Cancellations Tools Software Engineer Co-op, Bose July 2024 – Dec 2024
Technologies – JavaScript, React, Node.js, WebSocket APIs, Git, Agile Development Framingham, MA

- Developed Bose Channel Monitor, a **React SPA** using **React Router** enabling navigation (monitoring, configuration) for multichannel audio visualization, reducing workflow setup time by **60%** for **40+ engineers**
- Engineered **WebSocket client** handling 48kHz audio streams via **ArrayBuffer**, achieving **sub-10ms** low-latency
- Created high-performance UI components using **Chart.js** rendering real-time waveforms and spectrum analyzers, implementing **React.memo optimization** and AI tools (Co-Pilot, Cursor, GPT-4) for enhanced modularity
- Overcame performance bottlenecks by implementing **lazy loading**, **memoization**, and batched state updates, achieving **92%** performance improvement and reducing test cycles from **45 to 4 minutes**
- Enhanced testing flexibility by designing responsive CSS Grid dashboard with draggable tiles and custom **React hooks** for DSP operations using **Web Workers** for FFT computations to prevent main thread blocking

Software Engineer, Ernst & Young July 2022 – Aug 2023
Technologies – React, React Libraries, JavaScript, Node.js, Express, Redux, REST, SQL Bengaluru, India

- Led the migration of critical **SEO DIR pages** (B2B client's supplier listing platform) to **React & Node.js** setup with **SSR** and **CSR** hybrid architecture, ensuring reduced **load times (-17%)** & improved search rankings
- Implemented the CityBar feature, used for instant supplier lookup according to cities using **React Router** and **Redux**, reducing navigation **latency by 165ms** and ensuring seamless **SPA** transitions for improved UX
- Engineered scalable **REST API endpoints** for DIR Listings with **pagination** and lazy loading, enhancing **React hydration** and reducing payload size with progressive data fetching
- Developed & Integrated UI for **PhotoSearch**, an enterprise AI-powered visual search engine, using React, Redux, & **virtualized rendering**, improving real-time search experience, leading to **12% increase** in user engagement
- Received the **Excellence of Service** award at Ernst & Young for outstanding contributions in developing innovative and high-quality software solutions

Software Analyst Intern, Ernst & Young March 2022 – June 2022
Technologies – Python, Numpy, Keras, PyTorch, Pandas, Jupyter Notebook Bengaluru, India

- Pioneered a revenue assurance project to **reduce trade leakages** and proactively identify risks for a jewelry giant
- Developed custom **Python scripts** to automate monitoring, generating **real-time alerts** and updates, resulting in a **2.3X increase** in business efficiency compared to the traditional method
- Prototyped end-to-end anomaly detection solutions that reduced revenue leakage by **\$265K in one quarter**

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Java, SQL, HTML5/CSS3
Frontend	React.js, Redux Toolkit, Next.js, Tailwind CSS, Webpack, Jest, SASS
Backend & APIs	Node.js, Express.js, RESTful APIs, GraphQL, JWT Auth, WebSockets
Tools & Cloud	Git, Kibana, Postman, AWS, Docker
Software Engineering	System Design, Distributed Systems, Data structure & Algorithm

PROJECTS

FlixGPT | *React, React Libraries, Redux, Javascript, Tailwind, Firebase, OpenAI API, Gemini API* [Link](#)

- Built **GPT-powered movie recommendation platform** by integrating TMDB API and Gemini's API, delivering **personalized** suggestions based on user preferences and mood context
- Implemented secure **authentication** with Firebase, optimized state management with **Redux Toolkit**, and implemented a responsive UI for seamless user experience

Echoes All Too Well – Spatial Audio Engine | *C++, Dolby Atmos-style Audio, Multi-threading* [Link](#)

- Built immersive spatial audio engine transforming stereo audio into realistic **3D soundscapes** using C++
- Developed scalable real-time audio pipeline architecture with multi-threaded processing, achieving < 10ms latency through **efficient memory management** and **algorithmic optimization** techniques
- Applied advanced **data structure** optimization in C++ for efficient **audio buffer management** and time-series audio data processing, reducing computational overhead by **30%** while maintaining quality